MAGNETOM

System

Replacement Instructions

RF-System

Aera
Skyra
Avanto fit
Skyra fit
Prisma
Prisma fit

M7-000.841.14.20.02 | 11.21

25 Body Coil (BC) Aera 1.5T/ Skyra 3T

25.1 Requirements

25.1.1 Safety Information

⚠ **WARNING**

Contact with high voltage conducting gradient connections during BC-Tuning!

Risk of personal injury (e.g., due to electric shock)

Follow the described steps for your own safety.
Press the power-off push button at the Gradient Control Unit GSSU/DGSU.

Switch-off the circuit breaker F1/GPA before removing the magnet rear funnel for BC-Tuning.

Secure the circuit breakers against unintended switch-on.

Wait at least 20 minutes to ensure that the GPA power stage capacitors are completely discharged.

At the gradient connections, check for ZERO volts with an appropriate measurement device.

EGA15.134

⚠ **CAUTION**

The foamed plastic and protective film around the body resonator provides noise protection, do not remove or damage! BC components can be easily damaged if not handled properly.

Risk of component damage!

Never deposit the body resonator on a sharp-edged surface. Plastic parts and capacitors could be damaged.

Use the delivered service tool for BC lifting. Use the shipping crate to store the BC.

To store the BC temporarily, always use the 2 foam blocks delivered with the replacement support tools.

⚠ CAUTION

Heavy load! Weight of the BC is approximately 130 kg!

Risk of personal injury (e.g., intervertebral discs- / back injuries)!

To lift the BC from and up to the patient table, 3-4 persons are required.

Use/mount the lifting handles at the BC.

If a rear support is present, remove its upper part before starting to work (follow the procedure in this description).

When lifting, stand upright in front of the load to avoid injuries.

25.1.2 Estimated Completion Time and Number of Persons

i

The Optical Length Meter (OLM) inside the BC has been discontinued. Even if it is still present on the old BC, it is not reinstalled on the new one.
Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:
- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)
Schedule 30 min. additional time for the Update.

Work Steps	Time	Persons
Preparations	45 min.	1
Removal/ Installation	240 min.	2 i

Work Steps	Time	Persons
Startup/ TuneUp/ QA	60 min. Note: additional 30 min. to perform Update Instructions to deactivate OLM, see note above.	1
Final Steps	n.a.	n.a.
	Total = 5.5 hr. (6 hr. with UI)	

Tab. 67: Completion Time

25.1.3 Aids or Tools

- Non-magnetic tool kit
- Non-magnetic measuring tape or gauge
- Waterproof marker
- Non-magnetic spirit level
- Non-magnetic torque wrench 5-20 Nm (material number 83 95829)
- Accessory for torque wrench (material number 8396116)
- Electric screwdriver recommended (for opening and closing the shipping crate).
- Non-magnetic flat tool (for example, plastic ruler) to reattach the sound-absorbing foam pads.
- Only if applicable: Sylomer parts Aera / Skyra / Skyra fit (material number 10676588)
(These parts are included in the new replacement BC. They must be separately ordered just in case the old body coil should be reinstalled but decoupling foam elements are damaged and need replacement.)

i

Only in case your system has an Optical Length Meter (OLM) installed which is not yet deactivated:

Check / order material for the following Update Instructions:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

25.1.3.1 Handling Tools delivered with the replacement BC

- Slide support for removing/ installing the Body Coil
- 2 additional lifting handles (+ 2 handles installed)
- 2 foam blocks for temporary BC storage at the ground

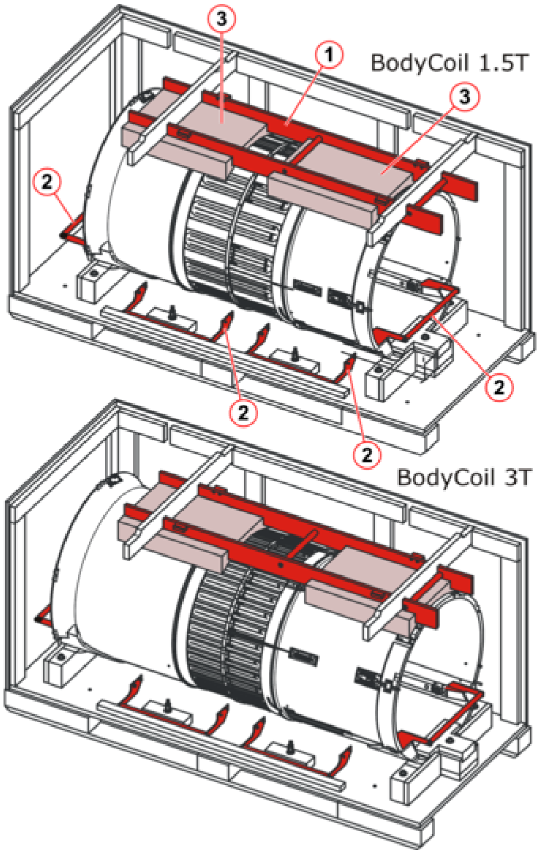


Fig. 225: Replacement BC on pallet with tools

1	Slide Support
2	Lifting handles
3	Foam blocks for temporary BC storage

BC Replacement Tool Set (10358867) delivered with the BC spare part:

Feeler gauge GC-BC098

3 distance cylinders BC098

2 distance markers

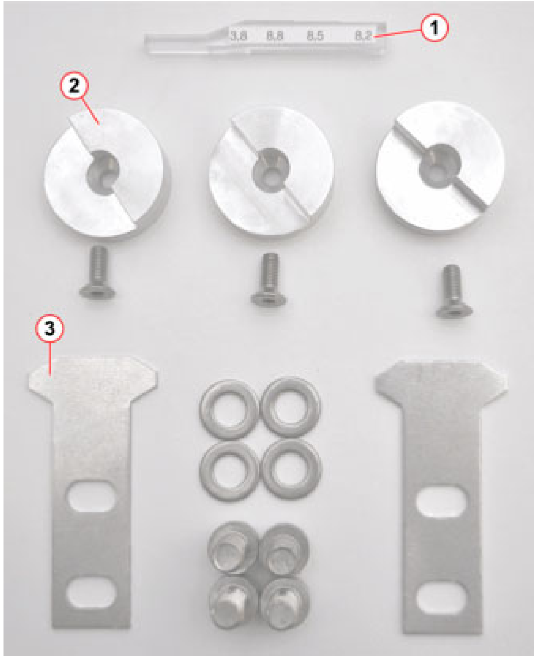


Fig. 226: Additional Tools

1 Feeler gauge GC-BC098 (10358339)

2 Distance cylinder BC098

3 Distance marker

The BC Replacement tool set for Aera/Skyra (10358867) is contained in the replacement BC or can be ordered individually from the Service Parts Catalog (SPC).

Set of small parts, delivered with the spare part BC:

- Screws and caps
- Nuts
- Washers
- Decoupling foam pads (Sylomer parts Aera / Skyra / Skyra fit, material number 10676588)
- Table end stop, table rail covers, and so on.

Fig. 227: BC small parts - example

25.2 Preparatory Work Steps

25.2.1 Magnet Cover parts and Power-off

i For details of disassembling/assembling cover parts, refer to the Replacement of Parts, System.
Aera / Skyra, "General Information".

Perform a QA/Phantom Shim Check, and check the actual shim situation.

If this is not possible, use the latest QA Phantom Shim report.

This step is necessary to verify, there is no shim deviation after BC replacement.
Remove the lower service cover at magnet.
Move the patient table fully down.
Remove the front funnel.
Remove the front funnel segment.
Select **System / Control / Meas Recon / Standby**. Secure the system against switch-on.
Switch-off F21/EPC and the F1 circuit breaker in the GPA.
After switching off power, wait at least 20 min. to discharge the GPA, then remove the rear funnel.
Remove the cover water hoses.

25.3 Removal

25.3.1 BC Rail Parts

The end parts of the BC rails must be removed to attach the delivered handle bars:

Remove the plastic caps with a small flat screwdriver.

Fig. 228: Removing plastic caps

Remove the rail end parts by releasing the attachment screws with a 5 mm Allen key.

Fig. 229: Removing rail end parts

On older systems, remove the fixation pin with a 6-mm wrench.

Fig. 230: Removing the securing pin (only for older systems)

For Skyra systems with the InSightec HIFU option, remove all option-specific BC rails. They are reinstalled on the new BC.

For details, see: **CB-DOC / Installation /Options / Side rail installation for InSightec HIFU system.**

BC Side Rails for InSightec HIFU systems

Fig. 231: Side rails, Skyra

25.3.2 Optical Length Meter OLM (if applicable)

The Optical Length Meter (OLM) inside the BC has been discontinued. Even if it is still present on the old BC, it is not reinstalled on the new one.

Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:

- MR005/15/P + MR019/15/R (Aera, Skyra)
- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

Schedule 30 min. additional time for the Update.

In case an OLM is installed at the system it must be removed from the BC:

Remove the plastic caps with a small flat screwdriver.

Release the OLM housing by removing 3 screws (Allen 5 mm).

Fig. 232: Removing the OLM

Loosen the screw in the back of the OLM housing.

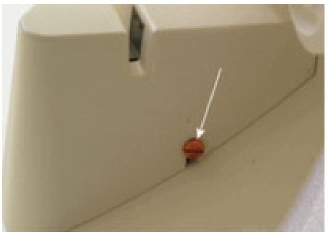


Fig. 233: Screw in back of the OLM housing

Disconnect and remove the OLM including FOCs from the BC.

Fig. 234: OLM and FOCs

i The replacement BC is equipped with a modified cover and blind caps to cover the cavities of the eliminated OLM.

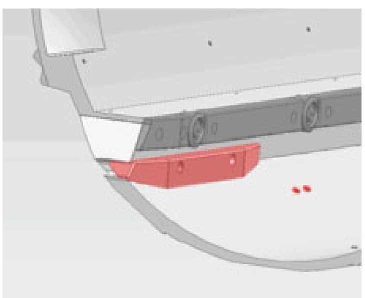


Fig. 235: Cover+blind caps for eliminated OLM - Aera+Sola Fit

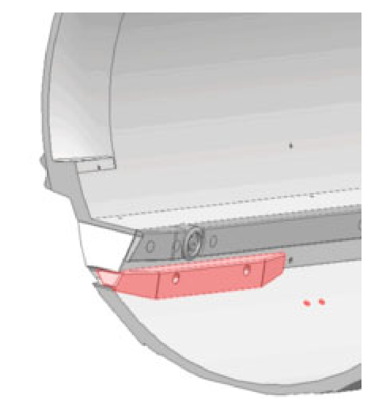


Fig. 236: Cover+blind caps for eliminated OLM - Skyra+Skyra fit+Vida Fit

25.3.3 Cable Connections

Disconnect the following cables at the front side of the BC:

Disconnect the 2 connectors to the bore lights on the front side (X3&X4) of the BC.

Fig. 237: Bore light connector front side

Disconnect the connector from the microphone board (on the BC front, 11 o’clock position).

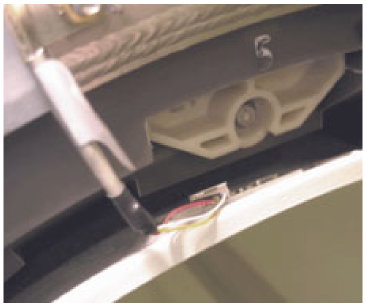


Fig. 238: Microphone connector - front-side

Disconnect the following cables on the rear side of the BC:

Disconnect the 2 connectors to the 2 bore lights (X1&X2).

Fig. 239: Bore light connector rear side

At the BC rear, pull out the microphone board (at 11 o’clock position) from its mounting bracket. Use a small, non-magnetic Allen key. Then, disconnect the connector from the microphone board.

Fig. 240: Pulling out the microphone board at BC rear side

Disconnect the connector from the ID-Prom board (at the 1 o’clock position). Use a small flat screwdriver to push down the latch (top center position) of the connector to release the lock mechanism.

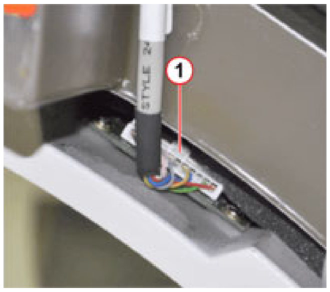


Fig. 241: ID-Prom board with connector

1 Push down the latch to release the connector

Disconnect the following cables of RF-components at the magnet:

Disconnect the 2 transmit cables at the TX-Box (BC_0 and BC_90).

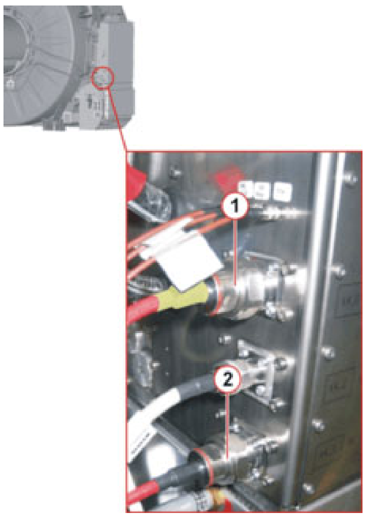


Fig. 242: Transmit cables at TX-Box - Example

1 TX-cable BC_0

2 TX-cable BC_90

Loosen the 3 screws and remove the shielded PIN diode supply cable at X8 of the BC_DYN in the RFCEL.

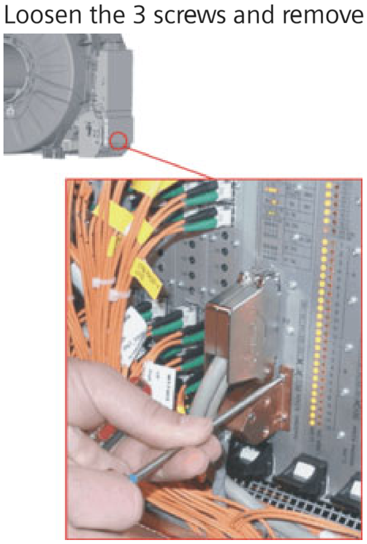


Fig. 243: Removing shielded connector X8 - BC_DYN

Loosen, but do not remove the ground clamps of the 2 red RF cables (1).

Loosen the gray PIN diode cable (2) at the magnet.

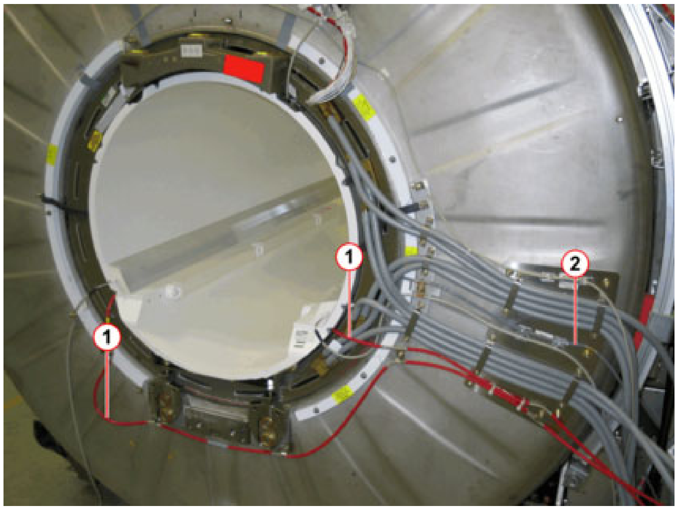


Fig. 244: RF-cabling in rear of magnet

1 RF-cables

2 PIN cable

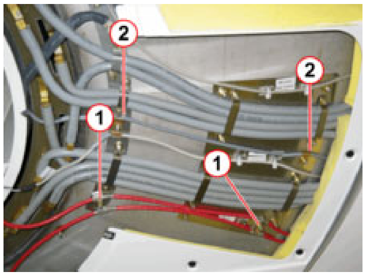


Fig. 245: RF cabling

1 Ground attachments of red RF cables

2 Ground attachments of grey PIN diode cable

Take the loose cables and roll them up to a loop. Then, place the cable bundle inside the BC tube and secure it with tape.

25.3.4 BC mechanical Removal

At the rear side of the old BC

Remove the thin, black, sound-absorbing foam between the BC and Gradient Coil.

Fig. 246: Foam removal

If a BC rear support is installed, its **upper** part has to be removed for better access and safe lifting of the BC.

⚠ CAUTION

Highly magnetic washers (labeled orange) are contained in the lower part of the rear support.

Risk of injury and component damage!

Never remove the lower part of the rear support under magnetic field!

ⓘ NOTICE

Attention, the upper part of the rear support contains slightly magnetic parts !

Risk of minor component damage if parts are slightly attracted by the magnetic field.

Hold slightly magnetic parts firmly with the hand and remove it immediately out of the magnetic field!

Mark the z-position of the upper part of the rear support relatively to the lower part at left and right side with a waterproof marker pen.

Loosen 4 nuts (2) (2 at left- and right-side) at the lower side of the upper part of rear support and remove it.

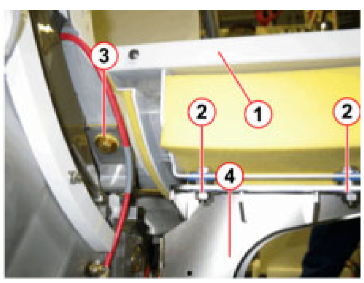


Fig. 247: Removeal of upper part of BC rear support

1	Upper part of BC rear support
2	Remove nuts
3	BC mounting screws (to be removed)
4	Lower part of BC rear support

Remove just the 2 BC screws (1), attaching the coil to the U-formed support bracket.

The BC support bracket itself, including nuts (2) and Allen screws (3), remain installed.

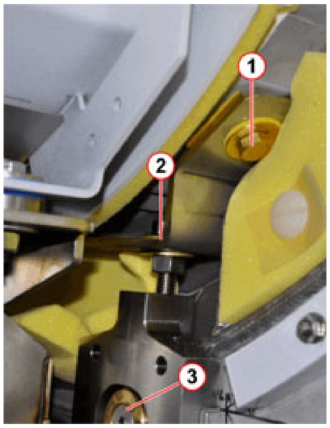


Fig. 248: BC support in rear (with installed rear support)

1 BC mounting screw (to be removed)

2 Adjustment nut and clamping nut (remain fixed)

3 Allen screw (6mm) at support (remains installed)

At the front side of the old BC

Attach the 2 X-position markers to the magnet with the delivered screws (to capture the horizontal X-position reference for the later BC reinstallation).

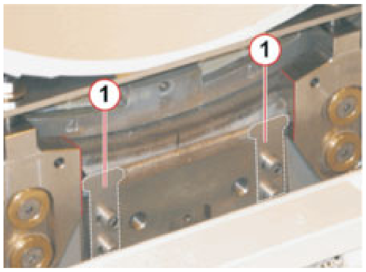


Fig. 249: Mounted X-position narkers

Mount the 2 additionally delivered handles to the front and rear of the BC.

Fig. 250: Mounted BC-handle

Disassemble the U-shaped BC support (1) completely by removing the 2 BC screws (2) and the 4 BC support screws (3) at the magnet.

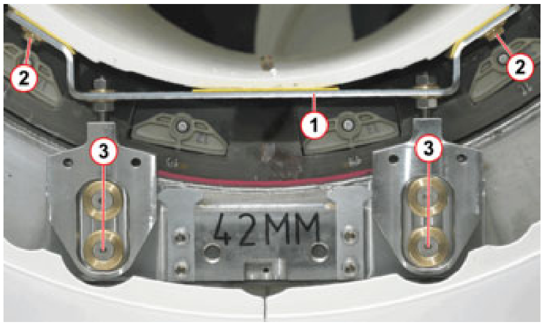


Fig. 251: BC front support

1 BC support

2 BC mounting screw

3 BC support screw

i To access the BC support screws, lift the patient table fully or undock it.

i Do not manipulate the BC height adjustment nuts. They remain fixed to ensure the correct reassembling height for the new BC.

Position the slide support on the patient table and adjust its height with reference to the BC.

Fig. 252: Slide support on patient table

1 Slide support

Adjusting the vertical table height for BC removal:

For a mobile table, use the foot pedals.

For the fixed table, engage the manual pump with the service lever (stored under the service access door of the foot cover).
The height of the slide support rails must align with the BC support guide rails.

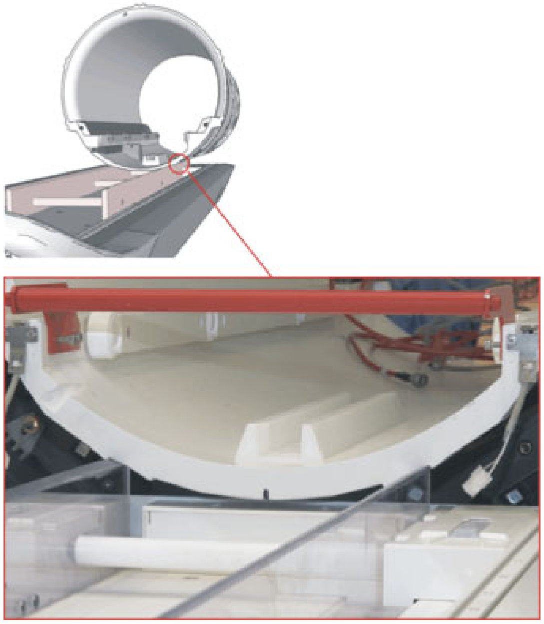


Fig. 253: BC support guide rails

Remove the old BC:

If your system has a VLM aspirating nozzle at the BC, carefully pull out the nozzle assembly from the BC mounting bracket.

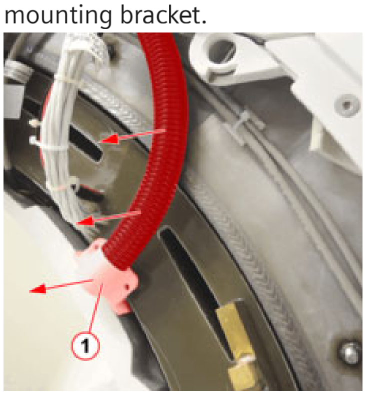


Fig. 254: VLM aspirating nozzle at BC - Example

1 Pull out nozzle assembly

NOTICE

The hose is glued to the nozzle assembly.

Risk of damage!

Pull out the complete nozzle assembly. Do not separate the hose!

2 persons are required for the following working steps.

⚠ CAUTION

Moving the heavy Body Coil.

Risk of personal injury ! Squeezing or cutting fingers and hand- / back injuries.

Use protective gloves.

Watch your fingers when moving the heavy BC.

Make sure, that the BC studs fully engage into the GC slide rails.

If a rear support is present, remove its upper part before starting to work (follow the procedure in this description).

When lifting, stand upright in front of the load to avoid intervertebral discs- / back injuries.

At the rear of the magnet, lift the BC a little bit from its support and carefully push the BC approximately 5 to 10 cm toward the front of the magnet.

Fig. 255: Lift up and move BC

Carefully lift-off the BC (rear-side) from the mounting support brackets, push it further to magnet front-side and make sure it is properly engaged to the GC slide rails.

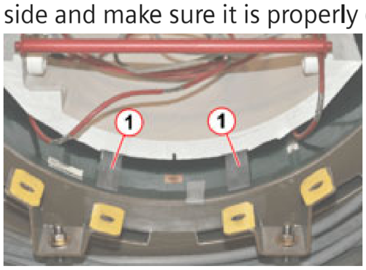


Fig. 256: BC on slide rails in the GC

1 GC slide rails

Slowly pull out the BC from the GC towards magnet front side. The guide studs of the BC must latch accurately into the support guide rails.

If necessary, readjust the patient table vertical height for a smooth transfer.

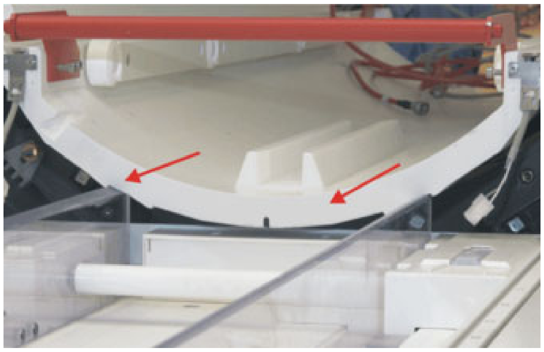


Fig. 257: Pull out BC from GC

i

Remove the old BC carefully to avoid damage to the Faraday Shield inside the Gradient Coil.

Carefully monitor the proper BC centering inside the GC.

Lift the BC (3-4 persons) off the patient table.

Fig. 258: BC pulled out from gradient coil on the slide support

Temporarily store the old BC on top of 2 foam support cushions delivered with the BC.

25.3.5 Exchange of Attachments in the BC Bore

Microphone board front-side

Use a small, non-magnetic Allen key, and insert its short end in the hole of the microphone board. Pull the PCB out of the clamping bracket.

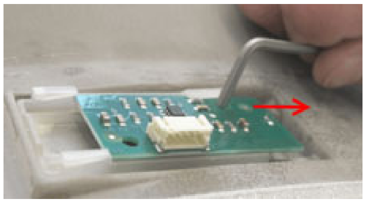


Fig. 259: Removal of microphone board

Attach the microphone board to the new BC in reverse order of removal.

Microphone board rear-side

Reattach the rear microphone board of the old BC to the new BC in the same fashion as before.

Bore lights

Remove left and right bore lights by removing 5 plastic screws each (flat screwdriver).

Fig. 260: Removing the bore light

i Plastic screws are fragile, do not overturn.

Remove the 2 attachment rails on each side of the BC and mount the rails to the new Body Coil (flat screwdriver).

Fig. 261: Removing the bore light mounting rails

Reattach the 2 bore lights to the new BC.

Chain Guide

Remove the chain guide from the old BC (two 4 mm Allen screws) and reattach it to the new body coil.

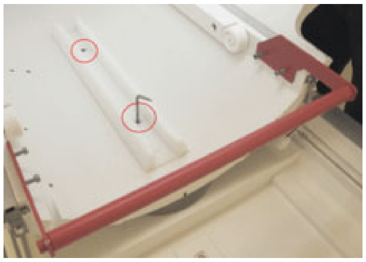


Fig. 262: Assembling the Chain Guide

Table End Stop

Remove the table end stop from the old BC (two 4 mm Allen screws) and reattach it to the new body coil.

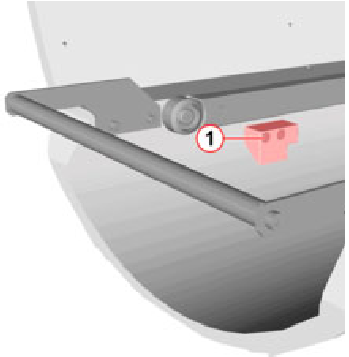


Fig. 263: Body Coil - service end

1 Table End Stop

i For systems equipped with the Whole Body option, the table end stop is mounted at the rear-support.

25.4 Installation

25.4.1 BC transfer into bore

Place the new BC on the slide support rails at the patient table using the lifting handles. The guide studs of the BC must accurately latch in the rails.

Move the BC forward to the Gradient Coil and adjust the height of the patient table. The height of the slide support rails must align with GC slide rails.

i Remove the old BC carefully to avoid damage to the Faraday Shield inside the Gradient Coil.

Carefully monitor the proper BC centering inside the GC.

⚠ CAUTION

Placing the heavy Body Coil on the supports.

Risk of personal injury (e.g., jamming of hands)!

Use protective gloves. Slowly move the BC on its studs at the GC slide rails.

i 3 to 4 persons are required for the following working steps.

Slide the BC carefully into the Gradient Coil.
A second person at the rear of the magnet has to ensure smooth transfer of the components to avoid mechanical damage to the BC and the Gradient Coil.

Stop approximately 10 cm before the final position and lift the BC carefully onto the rear mounting support brackets. Secure the BC with its 4 screws (hand-tight).

At the front, lift the BC approximately 20 mm to have enough clearance for mounting the complete BC-supports underneath. For lifting, use the provided BC-handles.

Fig. 264: Mounting the BC supports

Attach the mounting supports with four 6-mm Allen screws at the magnet front. Adjust the correct position with the previously established X-position markers; see Fig. 249.

At magnet front- and rear-side, fasten the BC to the U-formed support bracket with 2 screws each (hand-tight).

If your system has a VLM aspirating nozzle at the BC, carefully push the nozzle assembly into the BC mounting bracket.

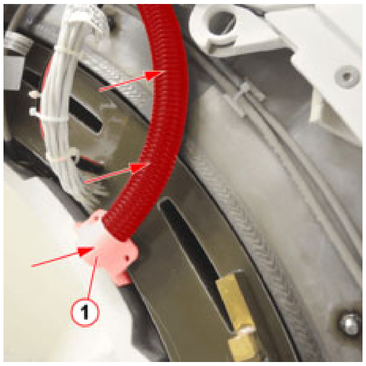


Fig. 265: VLM aspirating nozzle at BC - Example

1 Push-in nozzle assembly

i 2 persons are required for the following working steps.

⚠ CAUTION

Moving the heavy Body Coil.

Risk of personal injury ! Squeezing or cutting fingers and hand- / back injuries.

Use protective gloves.

Watch your fingers when moving the heavy BC.

Make sure, that the BC studs fully engage into the GC slide rails.

If a rear support is present, remove its upper part before starting to work (follow the procedure in this description).

When lifting, stand upright in front of the load to avoid intervertebral discs- / back injuries.

Carefully check all Sylodyn BC decoupling foam parts:

Carefully check that all Sylodyn foam parts Fig. 266 are undamaged and are mounted in a correct fashion.

Replace any damaged Sylodyn foam with parts contained in the kit.

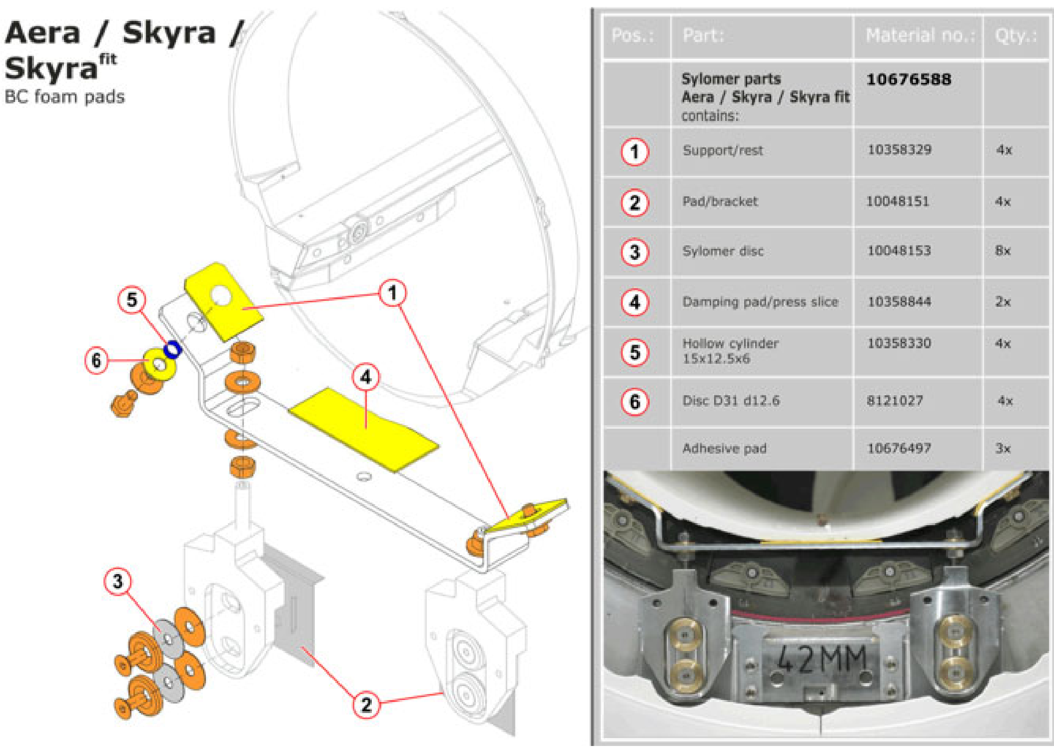
NOTICE

Incorrectly assembled or damaged Sylodyn BC decoupling foam parts.

If Sylodyn parts are damaged or incorrectly assembled, the BC mechanical decoupling has deteriorated. This may lead to image quality problems such as signal voids on diffusion-weighted images (DTI) or spikes.

Ensure all necessary Sylodyn foam formed parts are undamaged and mounted in a correct fashion.

i The Sylomer parts Aera/Skyra/Skyra fit are contained in the new replacement BC or can be separately ordered from the SPC (material number 106 76 588).



Pos.:	Part:	Material no.:	Qty.:
	Sylomer parts Aera / Skyra / Skyra fit contains:	**10676588**	
1	Support/rest	10358329	4x
2	Pad/bracket	10048151	4x
3	Sylomer disc	10048153	8x
4	Damping pad/press slice	10358844	2x
5	Hollow cylinder 15x12.5x6	10358330	4x
6	Disc D31 d12.6	8121027	4x
	Adhesive pad	10676497	3x

Fig. 266: BC decoupling foam parts

25.4.2 Mechanical BC Adjustment

Mount the 3 distance cylinders at the 4, 8 and 12 o'clock position to the BC.

Fig. 267: Mounted distance cylinder at gradient coil (aligned for 1.5T systems)

Adjust the BC in the Z-direction:

The distance cylinders can be rotated to the set the different BC z-position adjust length required for 3T or 1.5T systems.

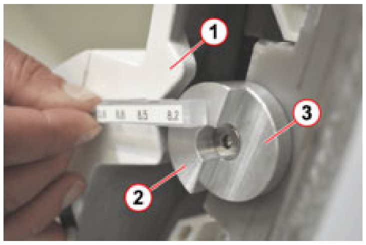


Fig. 268: Front surface BC

1 Front surface BC

2 Z-distance alignment for 3T

3 Z-distance alignment for 1.5T

Fig. 269: Aligning the z-distance of the BC (for 1.5T)

The adjustment is correct if the BC-front surface aligns with the corresponding top of the distance cylinder:

Distance for 3 T systems: 16 mm to Gradient Coil edge surface.

Distance for 1.5 T systems: 8 mm to Gradient Coil edge surface.

Adjustment in the X- and Y-direction:

The Body Coil has to be radially centered at both ends.

The distance between the Body Coil and the Gradient Coil must be equal all around the whole circumference (tolerance +1/-1 mm).

On the front

Measure the 2 spaces between the BC and each distance cylinder (mounted at the 4, 8 and 12 o'clock positions). Check the distance with the small end of the BC feeler gauge and ensure that all spaces are equal.

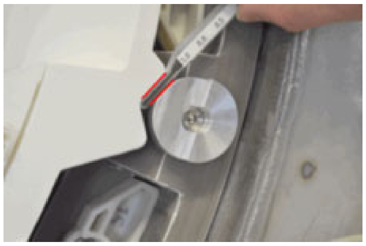


Fig. 270: Checking the distance on the upper side

Fig. 271: Checking the distance at the lower side

At the rear side

Measure at the 4, 8, and 12 o'clock positions using the BC feeler gauge. Insert the gauge into the gap between the BC and GC.

Fig. 272: Feeler gauge GC-BC098

1 Distance measurement steps

Fig. 273: Measuring the radial distance between BC and GC

i For measurement, the feeler gauge side with increments must point toward the inner diameter of the Gradient Coil.

Use a non-magnetic spirit level to align the Body Coil horizontally in the X-direction.

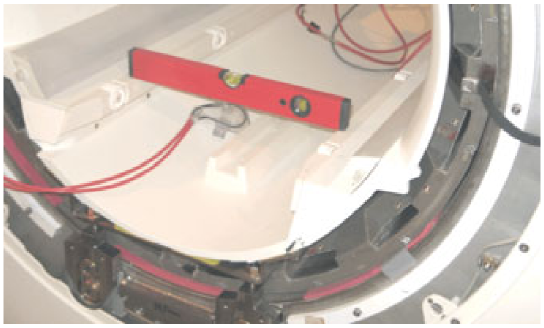


Fig. 274: Aligning the X-direction with a spirit level

Adjust BC using the **adjustment nuts (1)** at the support brackets.

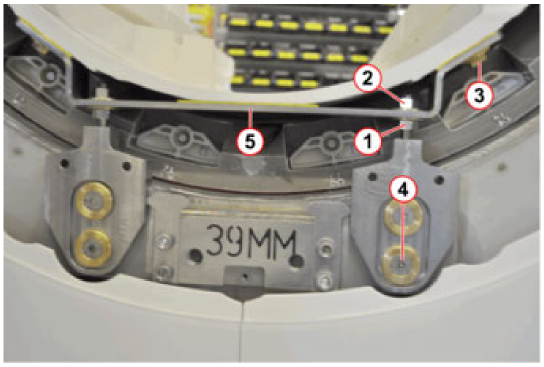


Fig. 275: BC support - front-side

1 Adjustment nut
2 Clamping nut
3 BC mounting screw
4 Allen screw at support
5 U-shaped mounting support bracket

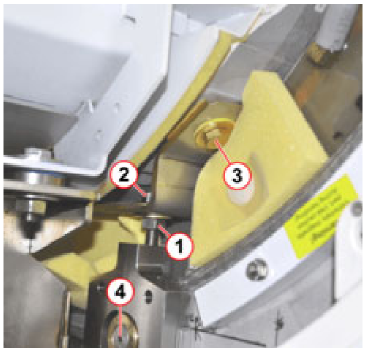


Fig. 276: BC support - rear-side

1 Adjustment nut
2 Clamping nut
3 BC mounting screw
4 Allen screw at support

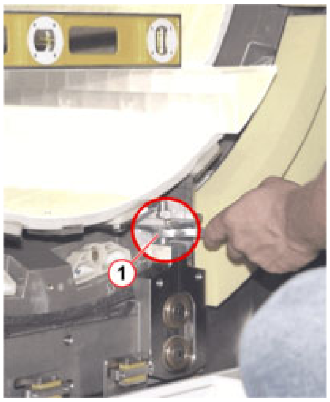


Fig. 277: Adjusting BC using adjustment nut at BC support

1 Adjustment nut

If all measured distances are in specification:

Tighten all screws and nuts of the coil supports:

Tighten the BC screws.

Check the distances again; if not OK, repeat the adjustments

25.4.3 Installation of BC Cables and grounding

Dismount the lifting handles.

Install the removed upper part of the rear support (if existing):

Attention weak magnetic object !

The space between the rear support and BC is 4 mm to allow proper attachment of the seal.

Route the RF cables from the BC back to the TX-Box.

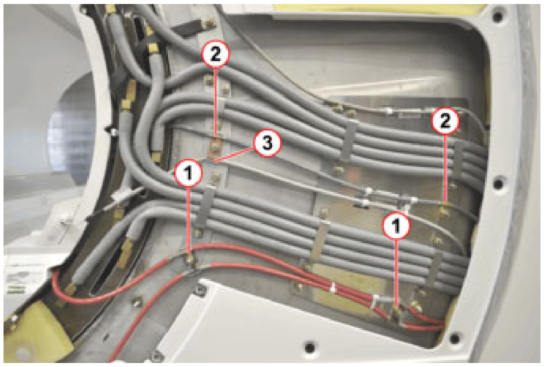


Fig. 278: Cabling magnet rear - Grounding points

1 Grounding points RF cables

2 Grounding points- Pin diode cable

3 Grounding point - bore light

Route and connect the PIN diode cable back from BC to the RFCEL/ BC_DYN X8.

Reconnect the cables to the bore lights, light marker, both microphones, and the EEPROM board of the BC. Make sure that the cables are properly routed. If necessary, secure them with cable ties.

Secure the RF cables carefully with clamps at the defined ground points of the magnet.

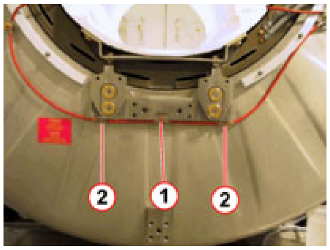


Fig. 279: Grounding point with/without rear support

1 One grounding point without rear support

2 Two grounding points with rear support

i For systems with a rear support, the RF cable BC_0 must be fed through the space between the rear support and magnet.

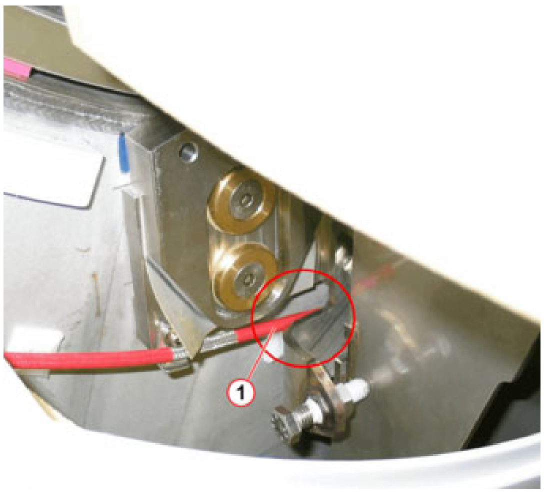


Fig. 280: RF cable routing with rear support

1 Feed cable BC_0 through gap between rear support and magnet

Tighten the clamps properly to avoid cable movement and spikes.

Fig. 281: Ground connection clamps - RF-cables

i To prevent spikes, insulate unused ground shields with tape.

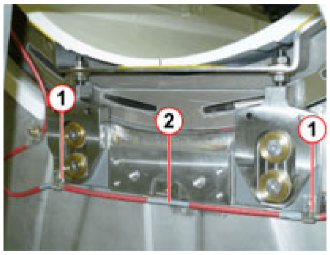


Fig. 282: RF cable insulations

1 Used ground point

2 Unused ground shield - insulated with adhesive tape

There are 2 additional RF-cable clamps at newer Aera systems:

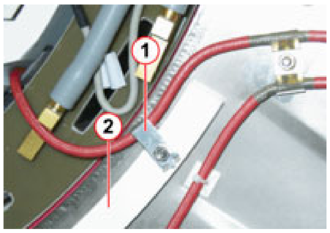


Fig. 283: Additional mounting clamp with Z2-shim

1 RF cable mounting clamp

2 Z2-shim segment

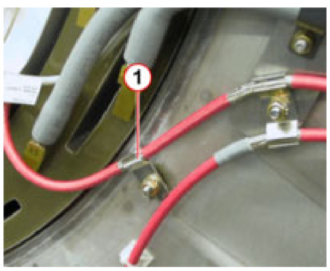


Fig. 284: Additional mounting clamp without Z2-shim

1 RF cable mounting clamp

Reconnect the RF-cables at the TX-Box (BC_0/BC_90).

Observe the specified torque for the RF-connectors at the TX-Box (see label at TX-Box):

TX_Box MR_Frequency	X4_0 BC 0	X4_1 BC 90	X4_2 LC
63.6 MHz	1.5 Nm (1_2 Nm)	1.5 Nm (1_2 Nm)	1.5 Nm (1_2 Nm)
123.2 MHz	15 Nm (10_20 Nm)	15 Nm (10_20 Nm)	1.5 Nm (1_2 Nm)

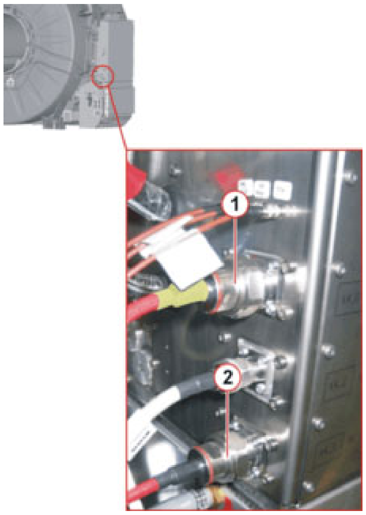


Fig. 285: Transmit cables at TX-Box - Example

1 TX-cable BC_0

2 TX-cable BC_90

25.4.4 Completion of Mechanical Parts

For Skyra systems with the InSightec HIFU option:

i

Reinstall all option-specific BC rails from the old BC to the new BC.

Adjust and check the patient table extraction force; see: **CB-DOC / Installation /Options / Side rail installation for InSightec HIFU system.**

BC Side Rails for InSightec HIFU systems

Fig. 286: Side rails, Skyra

Reattach the BC rail parts inside the bore.

Reattach the black, sound-absorbing foam in the space between the BC and GC (only at the magnet rear). Use a flat, non-magnetic tool (for example, a stainless steel or plastic ruler) to fit the foam strip into the space.

Foam insertion procedure (magnet rear-side only):

a) Press the flat tool against the leading edge of the foam and insert the strip carefully into the space between the GC and BC.

Fig. 287: Correct fitting - foam inserted at the leading edge first

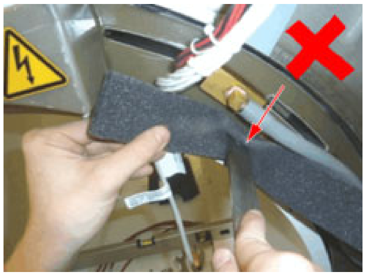


Fig. 288: Incorrect fitting - foam clumped together

NOTICE

Incorrectly fitted foam can lead to image quality problems.

Foam pads that are twisted or clumped together can couple mechanical vibrations from the GC to the BC, leading to signal voids on diffusion-weighted images (DTI).

Insert the foam pads smoothly, do not twist or clump together!

b) Reposition the flat tool to the middle of the strip and continue inserting the foam.

Fig. 289: Push the strip inside - tool in middle of the foam

c) Reposition the tool to the outer edge of the strip inserting the foam further inside.

Fig. 290: Push the strip farther in - tool at outer edge of the foam

d) Continue until the foam strip fully aligns with the edge of the Gradient Coil.

Fig. 291: Foam strip fully inserted

e) Continue to insert the foam strip along the circumference of the BC.

Fig. 292: Inserting the foam along the BC circumference

f) Here, at the 12 o’clock position, the first strip ends; continue with a new one. Make sure that there is no space left between ascending strips. This measure avoids acoustic leaks.

Fig. 293: 12 o'clock end of first strip - continue with the next one

i Leave no space between ascending foam strips. This measure avoids acoustic leaks.

g) Finish the insertion of foam strips along the whole circumference of the BC.

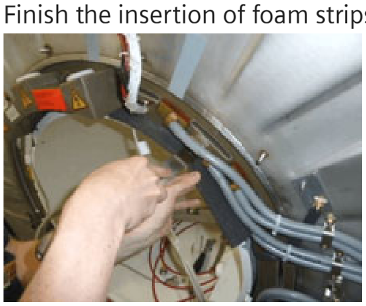


Fig. 294: Finish insertion of foam strips

h) Cut the last strip to the required length.

Fig. 295: Cut the last foam strip to required length

1 Excessive foam length - needs to be cut

i) The picture below shows a correctly inserted foam which aligns to the GC edge.

Fig. 296: Finished - correctly inserted foam strip

Reinstall the foam blocks at the magnet rear and front.

Fig. 297: Foam blocks rear-side

Fig. 298: Foam blocks front-side

Reassemble the system covers, except the rear funnel (required for the following BC tuning adjustment).

25.5 Start-up

i

The Optical Length Meter (OLM) inside the BC has been discontinued. Even if it is still present on the old BC, it is not reinstalled on the new one.
Check / perform "Deactivation of OLM" in the following Update Instructions together with a BC replacement:
- MR005/15/P + MR019/15/R (Aera, Skyra)

- MR021/15/P (Skyra pTX)
- MR023/15/R (Aera 24Ch)

Schedule 30 min. additional time for the Update.

25.5.1 Mechanical Work Steps

Switch-on the circuit breaker F1/GPA and F21/EPC.

Switch the MR system ON at the Alarm Box.

Pack the old BC including all delivered special tools and unused material in the shipping crate. The bundled BC cables must be attached to the coil with adhesive tape.

25.5.2 Software

i Starting with software version VD13A, the new body coil must be registered under **Service Software/Coils**. The old body coil serial number must be deleted there.

If SW VD13A or higher is installed, the new body coil must be registered under **SeSo/Coils**; otherwise **NO TuneUp or measurement is possible**:

Start the Service Software.

Start **Coils** menu.

Select **Body Coil/Update**; the new body coil is visible now.

Select the old BC serial number, and click Remove to delete the old BC.

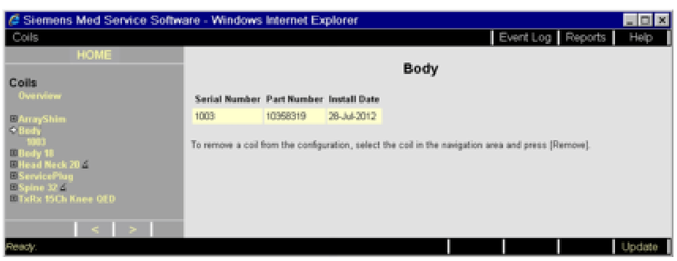


Fig. 299: SeSo>Coils Menu - Example

25.5.3 TuneUp/ Adjustment

i The "BC Tuning" measurement is performed with the large spherical phantom.
If the "BC Tuning" is out of specification, a manual readjustment of the BC trimmer capacitors is necessary.

Perform TU/ BC Tuning (refer also to Online Help).

Perform TU/ BC Power Losses.

Perform TU/ Image Brightness.

25.5.3.1 BC Tuning

For accessing the trimmer capacitors at the BC rear-side, the rear funnel cover must be removed.

⚠ WARNING

Contact with high voltage conducting gradient connections during BC-Tuning!

Risk of personal injury (e.g., due to electric shock)

Follow the described steps for your own safety.
Press the power-off push button at the Gradient Control Unit GSSU/DGSU.

Switch-off the circuit breaker F1/GPA before removing the magnet rear funnel for BC-Tuning.

Secure the circuit breakers against unintended switch-on.

Wait at least 20 minutes to ensure that the GPA power stage capacitors are completely discharged.

At the gradient connections, check for ZERO volts with an appropriate measurement device.

EGA15.134

i The BC trimmer capacitor positions are slightly different for standard systems or Skyra systems with "Tim TX True Shape" option installed (see following figures).

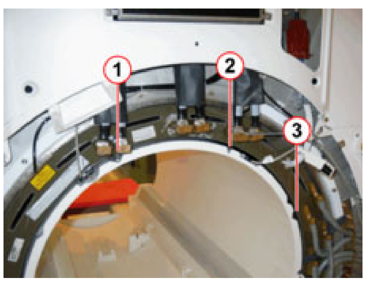


Fig. 300: BC trimmer capacitors rear-side (standard system)

1 Adjustment screw for trimmer T1 (BC2)

2 Adjustment screw for trimmer TD (Decoupling / Transmission)

3 Adjustment screw for trimmer T2 (BC)

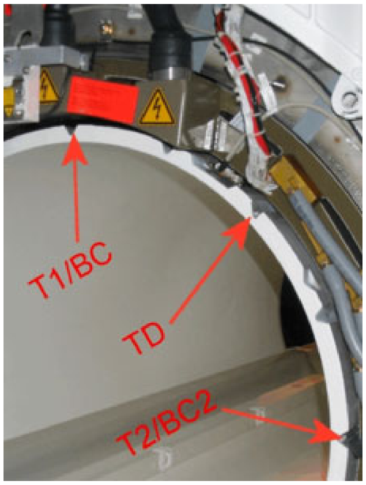


Fig. 301: BC trimmer capacitors rear-side (TIM TX True Shape option)

i For the BC Tuning adjust procedure, refer to the **Service Software/Online Help**.

Re-assemble the rear funnel cover.

25.5.4 Function Tests/Q Tests

Perform QA/ BC Tuning Check.

Perform QA/ BC Image Check.

Perform QA/ BC Power Losses Check.

Perform QA/ Spike Check.

Only for TimTX True Shape option: Perform QA/ PTX Image Check additionally.

Perform QA/ Phantom Shim Check:

Compare the new QA/ Phantom Shim results with the latest valid ones.

Compare the field terms, particularly term B(3,1) and term A(4,0).
If the difference is greater than 1 ppm or the QA is out of specification, then perform the following steps (otherwise stop here):

Remove the 4 inner guide rail screws from the new BC.

Use the slightly magnetic screws from the old BC and reestablish the old shim situation.

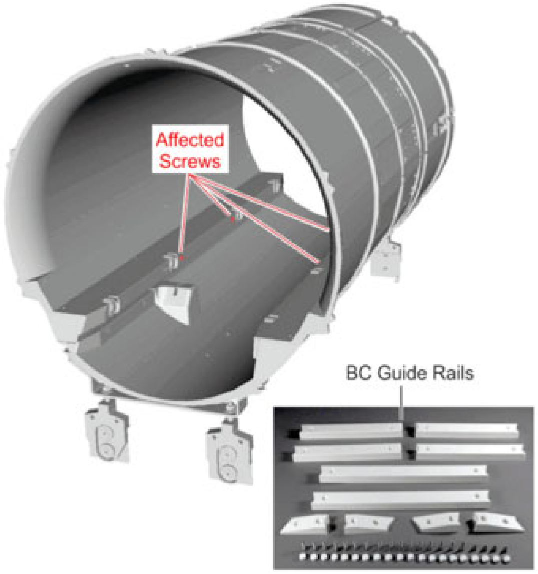


Fig. 302: BC guide rail Aera, Skyra

25.6 Final Steps

Ensure that the shim is in specification by running QA/ Phantom Shim Check.